Anabel Yong

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Summary

I work at the intersection of probabilistic machine learning and materials discovery, building uncertainty-aware, data-efficient models that accelerate the search for novel molecules and materials. The questions driving my research are:

- **Uncertainty & Out-of-distribution Discovery** - How do we know when to trust a model's predictions on unseen chemistry, and can that (mis)trust itself guide exploration toward genuinely novel discoveries?
- **Bayesian Decision-Making** - When experiments are expensive and objectives conflict, how should an agent use physical knowledge to find useful materials in as few trials as possible?
- **DFT-Coupled Surrogates** - How can fast probabilistic surrogates stay faithful to first-principles physics and reliably flag when they are about to fail?

Publications

- **Yong, A.**, Hart, M., Schwaller, P. (2026). MOFjson: An Explicit Topological Representation of Metal-Organic Frameworks for Language Models (Accepted as Poster, NeurIPS 2026 ML4Molecules) Link: openReview
- **Yong, A.**, Tripp, A., Hosseini-Gerami, L., Paige, B. (2025). A study of EHVI vs fixed scalarization for molecule design. (Accepted as Poster, NeurIPS 2025 AI for Science Workshop.) Link: arXiv
- Sun, H., **Yong, A.**, Gilly, L, Jin, F. (2025). Enhancing Instruction Following Capabilities in Seq2Seq Models: DoLA Adaptations for T5 (arXiv preprint) Link: arXiv

Education

PhD in Machine Learning for Materials Discovery, EPFL **August 2026 – present**
Advisors: Prof. Philippe Schwaller (Laboratory of Artificial Chemical Intelligence) & Prof. Michele Ceriotti (Laboratory of Computational Science and Modeling)
Modules: EE-556 Mathematics of Data: from Theory to Computation, MATH-500 Mathematical Aspects of Materials Modelling, CS-550 Formal Verification, EE-411 Fundamentals of Inference and Learning

MSc Computational Statistics & Machine Learning, UCL (Distinction) **Sept 2023 – Sept 2024**
Advisors: Prof. Brooks Paige, UCL & Dr. Austin Tripp, Cambridge Machine Learning Group
Modules: Probabilistic & Unsupervised Learning, Machine Learning Seminar, Numerical Optimization, Statistical Natural Language Processing, Supervised Learning (Statistical Learning Theory)

BSc Biochemistry (with Mathematics), University of Edinburgh **Oct 2019 – June 2023**
Advisor: Prof. Georg Kustatscher
Modules: Bioinformatics, Numerical Methods for Differential Equations, Linear Programming, Modelling & Solutions, Mathematics for Natural Sciences, Computer Simulation, Programming & Data Analysis

Research Experience

- **Graduate Student Researcher, National University of Singapore (PI: Prof. Jonathan Scarlett)** **May 2025 – Sept 2025**
  - Researched Bayesian optimization algorithms with diversity and safety constraints, focusing on uncertainty-aware batch selection for molecular design.
  - Benchmarked batch Bayesian optimization methods on multi-property optimization tasks, analyzing trade-offs between utility, Pareto front quality, and chemical diversity.

- **Graduate Student Researcher, IgnotaLabs.AI (PIs: Dr. Austin Tripp, Cambridge MLG Group, Layla Hosseini-Gerami & Prof. Brooks Paige, UCL)** **June 2024 – Sept 2024**
  - Built exact Gaussian process models on molecular fingerprint features for data-efficient multi-objective molecular design.

  - Evaluated Pareto-aware Bayesian optimization against fixed-weight scalarization, with better recovery of chemically diverse trade-off solutions.

- **Undergraduate Student Researcher, Kustatscher Lab, University of Edinburgh Oct 2022 – July 2023**
  - Benchmarked false discovery rate strategies for large-scale mass-spectrometry proteomics, improving confidence in protein and isoform identification.
  - Contributed to ProteomeHD2 curation by identifying previously uncharacterised novel microproteins, supporting expansion of the human proteome database.

- **Undergraduate Student Researcher, ZiHeng Yang Lab, UCL June 2022 – Sept 2022**
  - Implemented Bayesian MCMC methods to model ABO blood-group frequencies, using probabilistic inference to study genetically determined molecular phenotypes in human populations.

Skills

Programming:	Python, R, MATLAB, git, Linux, Google Cloud Platform (GCP), slurm, LaTeX
Machine Learning:	PyTorch, Jax, BoTorch, GPyTorch
Cheminformatics:	ChemDraw, RDKit, PyMOL, UniProt, MaxQuant, AutoDock Vina, BLAST

Awards & Scholarships

- **NCCR Separations PhD Doctoral Fellowship (2026)** - Fully funded by the Swiss National Science Foundation (SNSF) (4 years)
- **Yayasan Sarawak Tun Taib Postgraduate MSc Scholarship (2023)** - Fully funded Master's scholarship, awarded to 1 of ≈830 applicants (≈£50,000)
- **EPSRC Undergraduate Biochemistry Research Bursary**, University of Edinburgh (2021) - £2,500

Services (Reviewer)

- NeurIPS AI for Science Workshop (2025)
- NeurIPS ML4Molecules: Agentic Systems for Molecular Sciences Workshop (2026)

References

- **Brooks Paige, Associate Professor in Machine Learning, University College London:** b.paige@ucl.ac.uk
- **Jonathan Scarlett, Associate Professor in Dept. of Computer Science & Mathematics, National University of Singapore:** scarlett@comp.nus.edu.sg
- **Layla Hosseini-Gerami, Chief Data Science Officer, IgnotaLabs.ai:** layla.gerami@ignotalabs.ai
- **Austin Tripp, Senior Machine Learning Research Scientist, Valence Labs:** austin.james.tripp@gmail.com